Toronto Metropolitan University

Metropolitan Undergraduate Engineering Society

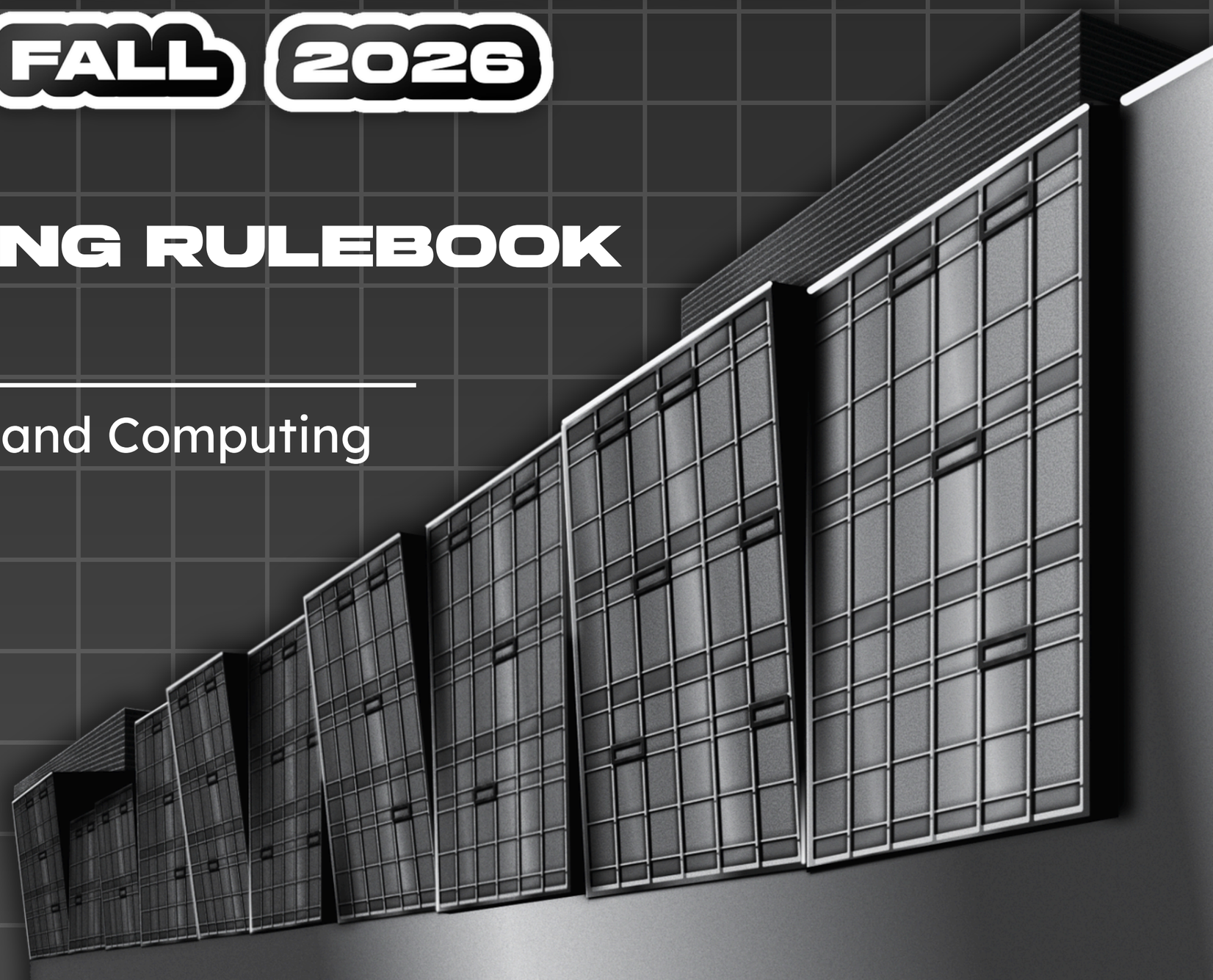
TORONTO METROPOLITAN UNIVERSITY ENGINEERING COMPETITION

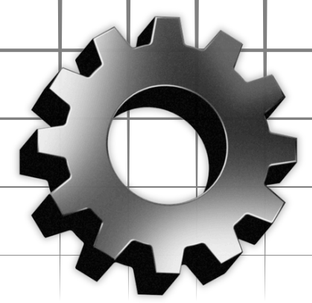
FALL 2026

RE-ENGINEERING RULEBOOK

NOVEMBER 7 & 8, 2026

George Vari Engineering and Computing Centre





1.1. OVERVIEW

Re-Engineering involves analyzing an existing engineering system, identifying its limitations, and proposing practical modifications to improve its performance. In MEC 2026, competitors will apply this process to food recovery and distribution systems, considering efficiency, reliability, sustainability, accessibility, and realistic implementation constraints.

an existing gadget or mechanism in order to add new functionality to it or enhance its original functionality. The intent of these improvements will be to extend the use of the device to an alternative situation or use case.

1.2 TEAM COMPOSITION

The Re-Engineering team will comprise a maximum of two (2) competitors.

1.3 PROMPT

It is recommended that the topic incorporate more than one engineering discipline. The topic should challenge competitors' technical knowledge and skills, it should also require competitors to evaluate the economic, environmental, political and social implications of their proposed solutions and address the requirements of the customer.Competition topics will be ones that could exist in the real world. Topics drawn from reality must be fully documented. All necessary documentation must be provided to the competitors when the problem is presented.

1.3.1 Cases

1.3.1. The competition will consist of two (2) sets of written cases.

1.3.2. Case 1 will be distributed to registered competitors six (6) days before the competition. Case 2 will be released during the competition briefing on Saturday, November 7, 2026.

1.3.3. The second case set will be presented to the competitors during the Competition Briefing

1.3.4. Both cases are due at the same time (the end of the competition phase).

1.3.5 The competition deliverables (report and presentation) must include both cases.

1.3.6. Case 1 Questions: Competitors may submit questions about Case 1 to the MEC Competition Coordinators at mec.vpcompetitions@mues.ca during the advance preparation period. Any clarifications affecting competition requirements will be communicated consistently to all registered teams.

1.3.6.1 Questions asked to the OEC OC within forty-eight (48) hours of the competition opening

1.4 LANGUAGE

The presentations and reports must be consistent in language.

1.5 RESOURCES

1.5.1 Facilities Required

- One (1) amphitheater (presentation of problem and team presentations)
- One (1) workroom per team (design of solution)

1.5.2 Personnel Required

1.5.2.1 Judges

A minimum of three (3) judges (and in any excess, an odd number of judges) are required to assess the problem-solving abilities, proposed solution, communication skills and team dynamics of the competitors. Judges in this category should come from a variety of backgrounds including communications, sales and technical or consulting engineering experience related to the topic.

1.5.2.2 Competition Leads

The Competition Leads are responsible, along with the VP Competitions, for the design and implementation of the Re-engineering competition. The Competition Leads must present the problem at the beginning of the competition and answer any questions raised by competitors. Only the Competition Leads may answer questions during the design period. The Competition Leads will also answer any questions the judges may have during the competition.

1.5.2.3 Official Timekeeper

The Official Timekeeper must be responsible for enforcing time limits during the question period, solution development and presentations, as stated in the rules.

1.5.3 Equipment Provided by MEC

The following equipment will be made available to all competitors during the competition time period:

1.5.3.1 During the design phase

- A design room with at least one (1) table, two (2) chairs and a whiteboard or blackboard
- One (1) digital storage device to submit the team presentation (USB)
- Internet connectivity
- Some information relevant to the design problem, at the discretion of the Competition Leads

1.5.3.2 During the presentation

- One (1) digital projector
- One (1) computer containing the team’s presentation file
- Whiteboard(s) or blackboard(s)

1.5.4 Allowed External Resources

The competitors are allowed to bring the following equipment with them:

- Any electronic material stored on CDs, USB keys, disks or other storage devices
    - Please check with the organizers to make sure the format of your electronic information will be accessible using the computers provided by OEC
- Any background research conducted by team members prior to the competition.
- Any textbooks, course notes or other reference material
- Each team member is allowed one computer

1.5.4.1. Teams are permitted to bring a slideshow template for use in the competition.

1.5.4.2. Templates may contain background images and a basic slide layout.

1.5.4.3 Slides are to have no specific content pre-competition.

Note: Since the use of the Internet and other external resources is permitted in this competition, all information used by competitors must be referenced very carefully. Competitors are not permitted to submit work completed by anyone other than the members of their team. If there is any evidence that competitors are submitting plagiarized work, the entire team will be eliminated from the competition and their home schools will be notified. Volunteers will monitor each team during the design process to deter teams from cheating, however, competitors are expected to act in good faith with the spirit of the competition.

1.6 PROCEDURE / TIMELINE

1.6.1 Case Structure

The competition will consist of two (2) written cases that will prompt the competitors to apply the re-engineering process and propose a solution.

1.6.2 Pre-Competition

Prior to the competition, information about the rules of the competition will be available on the competition website, metengcomp.com.

1.6.3 Competition

1.6.3.1 Question Period

Competitors then have fifteen (15) minutes to ask the Competition Leads any questions. Only the time used to ask the questions (not the answers) should be counted in the fifteen (15) minutes. Time used by judges to pose questions will not be counted.

1.6.3.2 Solution Development

Teams will be given eight (8) hours on Saturday, November 7, 2026, to complete their competition-day work, finalize both case solutions, and prepare their required deliverables. All deliverables shall be submitted to the competition director before the end of the allotted time. Competitors may finish before the end of the allotted time.

1.6.3.3 Rest Period

Competitors must be allowed a minimum of one (1) hour to rest before the presentation phase starts.

1.6.3.4 Solution Presentations

The presentation should cover both completed cases and should present both the material included in the reports and any additional information and explanations required in order to meet the evaluation criteria below. Competitors will have a maximum of fifteen (15) minutes to present their solutions. All team members must be present and participate in the presentation or be penalized by the judges. Judges then have a maximum of seven (7) minutes to ask questions. Judges can ask a question at any time during the presentation. The clock must be stopped during these interruptions. In order to ensure that all competitors cease to work on the case solutions once the design time has ended, the competitors cannot include any material in their oral presentation which is not included within their submitted written reports or presentation materials.

1.6.4 Timekeeping

The following rules will be adhered to with respect to timekeeping.

1.6.4.1 During the design phase

- Time is started when the teams leave the competition briefing
- The remaining time must be announced three (3) hours, one (1) hour, thirty (30) minutes, and ten (10) minutes before the end of the allotted time

1.6.4.2 During the presentation

- Time is halted when a judge asks a question during the presentation
- Time is halted when a team member answers a question asked by a judge
- The remaining time must be indicated to the competitors ten (10) minutes, five (5) minutes and one (1) minute before the end of the allotted time for the presentation to the judges
- A countdown presentation that is visible to both the judges and presenters must be given during the last thirty (30) seconds of the presentation

1.6.5 Presentation Order

- Presentation order shall be determined randomly
- Presentation order shall be announced thirty (30) minutes before the presentations commence. All teams are required to be present at this announcement
- Teams are not allowed to switch places in the presentation order
- Teams may not be in the audience for presentations of their competition until after they have completed their presentation. Any attempts at entering the presentation room before their allotted time will result in a warning for a first offense, and a ten (10) point penalty for any further offenses

1.6.6 Deliverables

1.6.6.1 For Competitors

Prior to the competition, information about the rules of the competition and the abstract will be available on the competition website, metengcomp.com.

1.6.6.2 From Competitors

Each team must electronically submit one combined final report covering both Case 1 and Case 2 before the end of the design period. The report must not exceed two (2) pages of main content, excluding the title page, references, and appendices. The report will be made available to the judges before each team's presentation. The report requirements will be outlined in the abstract. Each team is required to submit, electronically, before the end of the design period, a presentation about their proposed solution(s). The requirements of the presentation will be outlined in the abstract.

1.6.7 Response to Questions

- Only the competition director(s) may answer questions during the problem presentation and design phases. Volunteers and judges may not answer questions regarding rules and procedures at any time
- During the presentation of the problem, the Competition Leads shall answer orally and write down the answers provided
- During the design phase, only questions related to deliverable content shall be answered. No answers shall be provided in response to questions about the problem that might lead to the development of a new approach or which might invalidate a solution
- During the design phase, answers to questions shall be provided in writing to all teams at the same time. Fifteen (15) minutes may be allowed to elapse for a significant number of questions accumulated before responses are provided. A copy of the responses must be provided to the judges prior to the presentations
- During the week prior to the competition start time, all questions regarding the first case must be answered within forty-eight (48) hours

1.7 ASSESSMENT AND JUDGING

- The panel must have an odd number of judges
- The panel must have at least three (3) judges
- Feedback forms shall be provided to each team following the announcement of winners but prior to the end of MEC

1.7.1 Judging Matrix

Re-engineering Judging Matrix		
Solution	• Addresses Problem Statement • Design Feasibility • Environmental,Social, and Economic Analysis • Risk Analysis	Case 1 /25 Case 2 /35
Report	• Clarity /5 • Writing Style and Professionalism /5 • Deliverable Compliance with Expectations /5	/15
Presentation	• Voice, Articulation and Timing /10 • Visual Aids /10 • Response to Questions /5	/25
Penalties	• Plagiarism -50 • Insufficient Citation -50 • Documents Received After Deadline -50 • Absent Team Member -50 • Verbal Disclosure of School During Presentation -25 • Disclosure of School in Presentation Files/Documents -10 • Disclosure of School by Supporting Audience Members -10	-50 -50 -50 -50 -25 -10 -10
Total		/100